

INTERNSHIP PROJECT REPORT

PROJECT TITLE:DataEncryption and
Decryption tool.

INTERNSHIP NAME:CyberSecurity and
EthicalHacking.

Submitted by,

Susmitha C

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CERTIFICATE

This is to certify that the internship project titled 'Data Encryption and Decryption Tool' was successfully completed as part of the internship requirements.

ACKNOWLEDGEMENT

I express my sincere gratitude to my organization, mentor, faculty coordinator, and all those who supported me during the internship period.

ABSTRACT

The Data Encryption and Decryption Tool is developed to provide secure storage and transmission of information. The application uses AES and RSA cryptographic algorithms to encrypt and decrypt messages and files. The project demonstrates practical cybersecurity concepts including confidentiality, integrity, authentication, and secure communication. The system is designed with a user-friendly interface and can be used in educational, organizational, and commercial environments.

1. INTRODUCTION

Data security has become one of the most important concerns in the digital era. Organizations store and exchange large amounts of confidential information through networks and cloud platforms. Without proper security mechanisms, sensitive information may be exposed to unauthorized users. Encryption converts readable data into an unreadable form, while decryption restores it using authorized keys. This project provides a practical implementation of cryptographic techniques.

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1.1 Company/Internship Overview

During the internship, exposure was gained to cybersecurity principles, software development practices, secure coding techniques, and data protection mechanisms. The project was selected to understand how encryption technologies are applied in real-world environments.

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1.2 Problem Statement

Sensitive information transmitted through networks can be intercepted or modified by attackers. Traditional storage methods may not provide adequate protection. A secure system is required to encrypt and decrypt information efficiently.

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1.3 Objectives

To develop a secure encryption and decryption tool; implement AES and RSA algorithms; support text and file encryption; evaluate performance and security; and provide a user-friendly application.

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1.4 Scope

The project supports encryption and decryption of text messages and files. It can be used in educational institutions, businesses, healthcare organizations, banking systems, and cloud environments.

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2. LITERATURE SURVEY

Cryptography has evolved significantly over the years. Classical cryptography relied on substitution and transposition techniques. Modern cryptography uses mathematical algorithms such as AES, DES, RSA, ECC, and SHA. AES is widely adopted because of its efficiency and strong security, while RSA is commonly used for secure key exchange and authentication.

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2.1 AES Study

AES is a symmetric encryption algorithm standardized by NIST. It supports key sizes of 128, 192, and 256 bits. AES provides strong protection against brute-force attacks and is commonly used in banking, government, and cloud applications.

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2.2 RSA Study

RSA is an asymmetric encryption algorithm based on public and private keys. It relies on the mathematical difficulty of factoring large prime numbers. RSA is widely used for digital signatures and secure communication.

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3. SYSTEM ANALYSIS

System analysis focuses on identifying requirements and evaluating existing solutions. The proposed system addresses limitations of traditional methods by integrating modern encryption algorithms.

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3.1 Existing System

Data may be stored in plain text, making it vulnerable to unauthorized access. Existing solutions may be expensive, complex, or limited in functionality.

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3.2 Proposed System

The proposed system provides secure encryption and decryption using AES and RSA. It supports file handling, key management, and performance comparison.

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3.3 Advantages

Improved security, confidentiality, ease of use, portability, scalability, and support for multiple file formats.

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4. FEASIBILITY STUDY

The project is technically feasible using Python and open-source cryptography libraries. It is economically feasible due to low development cost and operationally feasible because users can easily interact with the application.

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is technically feasible using Python and open-source cryptography libraries. It is economically feasible due to low development cost and operationally feasible because users can easily interact with the application.

5. REQUIREMENTS SPECIFICATION

Hardware: Intel i3/i5 processor, 4 GB RAM, 500 MB storage. Software: Windows/Linux/macOS, Python, VS Code, Cryptography library, Tkinter GUI framework.

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5.1 Functional Requirements

User input handling, encryption, decryption, file selection, key generation, result display, and storage of encrypted data.

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5.2 Non-Functional Requirements

Security, reliability, maintainability, performance, usability, portability, and scalability.

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6. SYSTEM DESIGN

The application consists of Input Module, Encryption Module, Decryption Module, Key Management Module, File Handling Module, and Output Module.

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Module, Decryption Module, Key Management Module, File Handling Module, and Output Module.

6.1 Architecture

User → Input → Encryption Engine → Ciphertext Storage → Decryption Engine → Output.

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6.2 Module Description

The Input Module collects user data. The Encryption Module converts plaintext into ciphertext. The Decryption Module restores the original data. The File Module manages file operations.

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7. UML DIAGRAMS

Use Case Diagram identifies interactions between the user and the system. Class Diagram represents system classes. Sequence Diagram shows message flow. Activity Diagram represents workflow. These diagrams can be inserted during final formatting.

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8. METHODOLOGY

Requirement gathering, system design, algorithm selection, implementation, testing, validation, and documentation were performed sequentially.

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9. ALGORITHMS USED

AES uses symmetric encryption while RSA uses asymmetric encryption. Both algorithms contribute to secure communication and data protection.

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9.1 AES Working

Plaintext is divided into blocks and transformed through multiple rounds involving substitution, shifting, mixing, and key addition.

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9.2 RSA Working

Keys are generated using large prime numbers. Public keys encrypt data and private keys decrypt data.

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9.3 AES vs RSA

AES is faster and suitable for large files, whereas RSA is computationally intensive but ideal for secure key exchange.

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10. IMPLEMENTATION

The system was implemented using Python. Cryptographic functions were developed using standard libraries. A graphical interface allows users to select files, enter text, and perform encryption/decryption.

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10.1 Development Environment

Python, PyCryptodome/Cryptography library, Tkinter GUI, and Visual Studio Code.

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10.2 Sample Workflow

User enters text → selects AES/RSA → encryption performed → ciphertext displayed → decryption restores original message.

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11. TESTING

Testing ensures that all modules function correctly under different conditions.

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11.1 Unit Testing

Each module was tested independently.

Each module was tested independently. Each module was tested independently. Each module was tested independently.

11.2 Integration Testing

Interactions between modules were validated.

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11.3 System Testing

The entire system was tested as a complete application.

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11.4 Security Testing

Unauthorized access attempts were simulated to verify protection mechanisms.

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12. RESULTS AND DISCUSSION

The system successfully encrypted and decrypted text and files. AES demonstrated superior performance for large data sizes, while RSA provided secure key exchange capabilities.

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13. APPLICATIONS

Banking, cloud computing, healthcare, military systems, government services, e-commerce platforms, educational institutions, and enterprise security.

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14. CHALLENGES FACED

Understanding cryptographic concepts, key management, handling file encryption, testing security features, and optimizing performance were key challenges.

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15. LEARNING OUTCOMES

Understanding encryption principles, implementing AES and RSA, software development practices, cybersecurity concepts, and testing methodologies.

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16. FUTURE ENHANCEMENTS

Cloud integration, mobile application support, digital signatures, biometric authentication, blockchain integration, and multi-factor authentication.

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17. CONCLUSION

The internship project successfully demonstrated the implementation of a secure Data Encryption and Decryption Tool. The project provided practical exposure to cryptography, cybersecurity, and software development. The system effectively protects sensitive information and highlights the importance of secure communication in modern computing environments.

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